

Inline Assembly:

Write a C++ program that uses **inline assembly** to move the value of one 32-bit register into another register.

Declare an integer variable in C++ and use **inline assembly** to load its value into the **EAX register**.

Write a C++ program that uses inline assembly to **add two C++ integer variables** and store the result in another variable.

Write a C++ program that uses inline assembly to **increment all elements of an integer array** using a loop in assembly language.

Write an inline assembly code snippet that modifies the **EBX register** and then correctly **restores its original value** before exiting the assembly block.

Write a C++ program that:

- Declares and initializes an integer array of **5 elements**
- Uses **inline assembly** to **swap the least significant byte and most significant byte of each element**
- Prints the updated array

Write a C++ program that:

- Declares an integer variable
- Uses **inline assembly** to swap:
 - Byte #0 (LSB)
 - Byte #3 (MSB)
- Display the result

Write a C++ program that:

- Declares an integer array of **5 elements**
- Uses **inline assembly** to rotate **all bytes of each element to the left by 1 bit**
- Display the updated values

Write a C++ program that:

- Declares an array of **5 integers**
- Uses **inline assembly** to rotate:
 - o Even indexed elements to the **left**
 - o Odd indexed elements to the **right**
- Print the array

Write an **inline assembly function** that:

- Rotates an **integer value** either left or right
- Direction is passed as a parameter:
 - o 0 → rotate left
 - o 1 → rotate right
- Call this function from main() and display the result